

NeuroFleetX – Frontend Documentation

User Module

Introduction

The User module of NeuroFleetX is designed to provide a smooth and intuitive experience for passengers. It allows users to request rides, view analytics, and track their ride history through a clean and role-based frontend architecture.

Frontend Technology Overview

- React.js using functional components
- React Router DOM for navigation and routing
- Context API for authentication and user state management
- REST API integration with Spring Boot backend
- Protected routes to ensure role-based access

User Authentication & Access Flow

Users log in using their credentials. After successful authentication, the backend identifies the role as `ROLE_USER`. The frontend then redirects the user to the User Dashboard and enables access only to user-specific pages.

User Dashboard

The User Dashboard acts as the main entry point for the user. It provides navigation to key features such as New Ride Request, Ride History, Analytics, and Profile. The dashboard is designed to give a quick overview and easy access to all user functionalities.

New Ride Request Page (Core Functionality)

This page allows users to request a new ride. Users enter origin and destination coordinates along with traffic and weather information. The frontend sends this data to the backend to create a ride with `PENDING` status. Users can also predict ETA using AI-based services and assign the most suitable available taxi.

Ride History Page

The Ride History page displays all rides requested by the user along with their current status such as Pending, Assigned, or Completed. This information is fetched dynamically from the backend and helps users track their past activities.

User Analytics Page

The Analytics page provides summarized insights instead of raw ride records. It shows metrics such as total rides, completed rides, and average ETA. This helps users understand usage patterns and overall ride behavior.

User Profile Page

The Profile page displays personal user information such as name and email. It ensures identity visibility and supports future profile enhancement features.

Security & Role-Based Access Control

All user pages are protected using role-based routing. Only authenticated users with `ROLE_USER` can access these pages, and unauthorized access is blocked even through direct URLs.

Conclusion

The User module of NeuroFleetX focuses on simplicity, usability, and real-time interaction with backend services. It demonstrates a real-world ride-booking flow combined with analytics and secure access control, making it suitable for scalable applications.